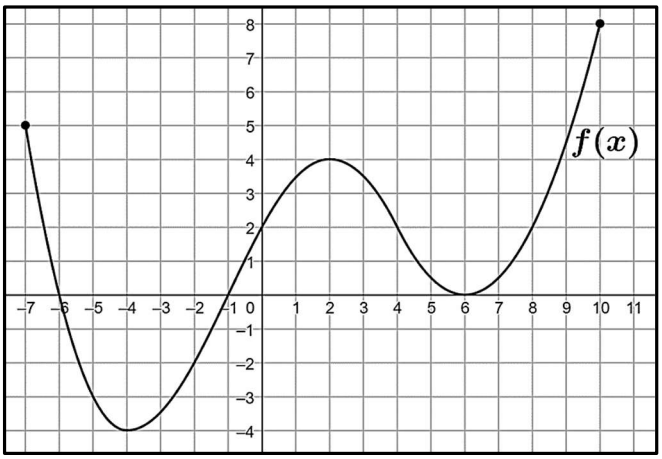
Worksheet 5-9 A Connecting f, f', and f'' #1
AP Calculus

Name SOLUTIONS

Directions: The graph of $f(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



1. At which x value(s) is $f(x) = 0$?

   $f(x) = 0$ at $x = -6$, $x = -1$, and $x = 6$

2. At which x value(s) is $f'(x) = 0$?

   $f'(x) = 0$ at $x = -4$, $x = 2$, and $x = 6$ because $f(x)$ has a horizontal tangent at these x-values.

3. On what interval(s) is $f(x)$ increasing?

   $f(x)$ is increasing on $[-4, 2]$, and $[6, 10]$

4. At which x value(s) does $f(x)$ have a relative minimum?

   $f(x)$ has a relative minimum at $x = -4$ and $x = 6$ because $f(x)$ changes from decreasing to increasing at these x-values

5. At which x value(s) does $f''(x) = 0$?

   $f''(x) = 0$ at $x = -1$ because $f(x)$ changes from concave up to concave down at this x-value and at $x = 4$ because $f(x)$ changes from concave down to concave up at this x-value.

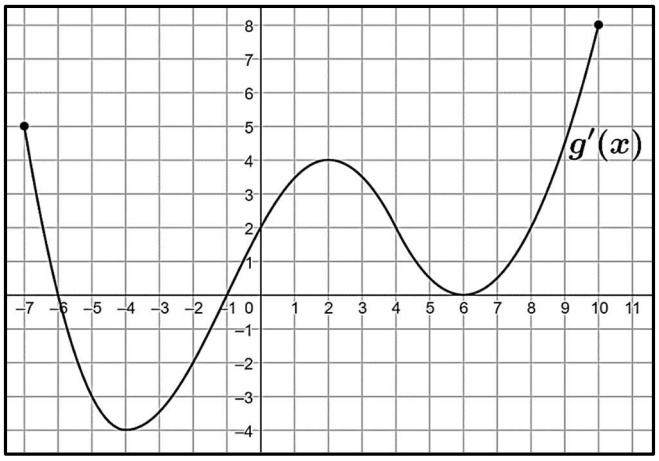
6. On what interval(s) is $f(x)$ concave down?

   $f(x)$ is concave down on $(-1, 4)$

7. At which x value(s) does $f(x)$ have a point of inflection?

   $f''(x) = 0$ at $x = -1$ because $f(x)$ changes from concave up to concave down at this x-value and at $x = 4$ because $f(x)$ changes from concave down to concave up at this x-value.

Directions: The graph of $g'(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



8. At which x value(s) is $g(x) = 0$? Explain your reasoning.

We cannot determine the x-values where $g(x) = 0$ because this is the graph of $g'(x)$.

9. At which x value(s) is $g'(x) = 0$?

$g'(x) = 0$ at $x = -6$, $x = -1$, and $x = 6$

10. On what interval(s) is $g(x)$ increasing? Explain your reasoning.

$g(x)$ is increasing on $[-7, -6]$ and $[-1, 10]$ because $g'(x) \geq 0$ on these intervals

11. At which x value(s) does $g(x)$ have a relative minimum? Explain your reasoning.

$g(x)$ has a relative minimum at $x = -1$ because $g'(-1) = 0$ and $g'(x)$ changes from negative to positive at $x = -1$

12. At which x value(s) does $g''(x) = 0$? Explain your reasoning.

$g''(x) = 0$ at $x = -4$, $x = 2$, and $x = 6$ because $g'(x)$ has relative extrema at these x-values (horizontal tangent lines).

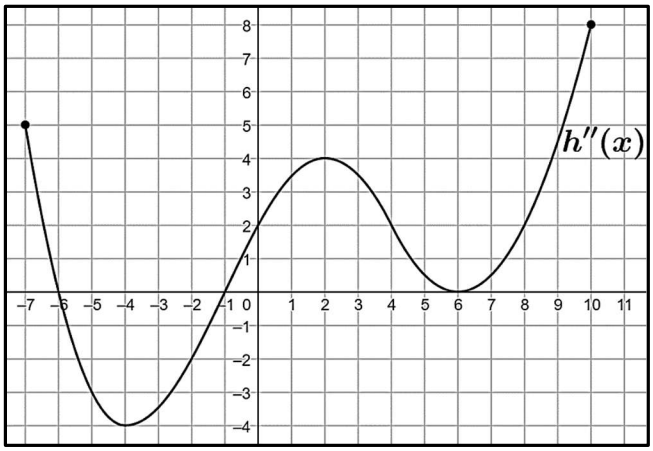
13. On what interval(s) is $g(x)$ concave down? Explain your reasoning.

$g(x)$ is concave down on $(-7, -4)$ and $(2, 6)$ because $g'(x)$ is decreasing on these intervals.

14. At which x value(s) does $g(x)$ have a point of inflection? Explain your reasoning.

$g(x)$ has a point of inflection at $x = -4$ and $x = 6$ because $g'(x)$ changes from decreasing to increasing at these points, and at $x = 2$ because $g'(x)$ changes from increasing to decreasing at this x-value.

Directions: The graph of $h''(x)$ is given below. Use the graph to answer the following questions. If not enough information is given to answer a question, write "Cannot be determined".



15. At which x value(s) is $h(x) = 0$? Explain your reasoning.

We cannot determine the x-values where $h(x) = 0$ because this is the graph of $h''(x)$, not $h(x)$.

16. At which x value(s) is $h'(x) = 0$? Explain your reasoning.

We cannot determine the x-values where $h'(x) = 0$ because this is the graph of $h''(x)$, not $h'(x)$.

17. On what interval(s) is $h(x)$ increasing? Explain your reasoning.

We cannot determine the x-values where $h(x)$ is increasing because this is the graph of $h''(x)$, not $h'(x)$.

18. At which x value(s) does $h(x)$ have a relative minimum? Explain your reasoning.

We cannot determine the x-values where $h(x)$ has a relative minimum because this is the graph of $h''(x)$, not $h'(x)$.

19. At which x value(s) does $h''(x) = 0$? Explain your reasoning.

$h''(x) = 0$ at $x = -6$, $x = -1$, and $x = 6$ because the graph of $h''(x)$ touches or crosses the x-axis at these x-values.

20. On what interval(s) is $h(x)$ concave down? Explain your reasoning.

$h(x)$ is concave down on $(-6, -1)$ because $h''(x) < 0$ on this interval.

21. At which x value(s) does $h(x)$ have a point of inflection? Explain your reasoning.

$h(x)$ has a point of inflection at $x = -6$ and $x = -1$ because $h''(x)$ changes signs at these x-values.

For each table, selected values of x and $f(x)$ are given. Assume that $f'(x)$ and $f''(x)$ do not change signs. Answer the questions for each table.

22.)

x	$f(x)$
4	-5
5	-8
6	-12
7	-17

f': -3, -4, -5

dec dec

a.) is $f(x)$ increasing or decreasing?

$f(x)$ is decreasing

b.) is $f(x)$ concave up or concave down?

$f(x)$ is concave down

23.)

x	$f(x)$
-3	-2
-2	3
-1	7
0	10

f': $+5$, $+4$, $+3$

inc dec

a.) is $f(x)$ increasing or decreasing?

$f(x)$ is increasing

b.) is $f(x)$ concave up or concave down?

$f(x)$ is concave down

24.)

x	$f(x)$
2	3
3	0
4	-2
5	-3

f': -3, -2, -1

dec inc

a.) is $f(x)$ increasing or decreasing?

$f(x)$ is decreasing

b.) is $f(x)$ concave up or concave down?

$f(x)$ is concave up

25.)

x	$f(x)$
0	1
1	2
2	4
3	7

f': $+1$, $+2$, $+3$

inc inc

a.) is $f(x)$ increasing or decreasing?

$f(x)$ is increasing

b.) is $f(x)$ concave up or concave down?

$f(x)$ is concave up